

HelixCode — Feature Status Summary

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Two-audience companion (§11.4.56) to `Status.md` for release `codeserver-1.0.0-dev-0.0.3`. Page 1 is for the project / product team; Page 2 is for software engineers. Always in sync with `Status.md` (§11.4.60).

Page 1 — For the project & product team

What HelixCode is. Browser-based VS Code that opens your real projects, behind a secure HTTPS front door, running without root — and now tied to your **real host account with your SSH key** (no password anywhere).

What works now (proven this release):

- **Log in with your SSH key, no password.** The login page shows a one-time challenge; you sign it locally with your key and paste it back. Wrong or missing signatures are refused. Proven end-to-end on the live system.
- **The editor is really you.** Once in, the editor runs as your host account, so your SSH keys, your `.bashrc` settings, and all your tools work in the terminal — including git over SSH. Proven live (a real git remote check returned results).
- **It stays locked if the guard goes down.** If the login guard stops, access is **denied**, never accidentally opened. Proven by killing the guard live and confirming access was blocked, then recovered.
- **Install and reboot survival.** One script installs everything without root; the services are set to come back after a reboot.

- **Plugins install from the marketplace.** A real editor extension was installed from the Open VSX marketplace and loaded — proven live, not just claimed. Popular plugins (Python, ESLint, Prettier, GitLens and more) were installed, stayed installed after a restart, and their settings round-tripped.
- **Dark editor by default.** Fresh installs now open in the "Visual Studio Dark" theme — proven not just in the config but by rendering the real editor and measuring the pixels (it came out genuinely dark, versus a light control for comparison).
- **Login-page copy/paste buttons work.** The one-click "copy the command" and "paste your signature" buttons on the login page are built, accessible, and proven by rendering the real page and driving the buttons.
- **Click-through marketplace test.** An automated browser opens the Extensions panel, searches the marketplace, and clicks the real Install button — the browse-and-click journey is proven (the final install-completion step still needs a person on a headless machine, honestly noted).
- **Secure, fast front door — now survives reboots.** HTTPS with a modern fast protocol (HTTP/3) and response compression; the plain-HTTP port redirects to HTTPS. The front door now runs as a boot-persistent service (rootless) and stays up across session crashes — proven live (running for over an hour, set to auto-start).

What is still in progress / needs the operator:

- **Watching yourself edit in the browser** (typing, clicking files) is confirmed **by a person** for now — the automated click-through is still being wired.
- **Real public HTTPS certificate** needs a public domain and is **enabled by the operator**; the machinery is ready.
- **A host firewall tweak** (a one-line rule) needs the operator to apply it as root — the project itself never uses root.
- **Feature videos.** We do not yet have recorded "watch it work" videos; every claim above is backed by a saved test-evidence file instead. Recording the videos is a tracked to-do.

Bottom line: the new real-account SSH-key login and the editor experience are **working and proven** on the live system (23 of 23 core suites passed, plus the post-0.0.3 additions — login copy/paste buttons, dark-by-default theme, popular plugins, the click-through marketplace journey, and the now-live boot-persistent front door — each proven with captured evidence). The remaining honestly-marked items are the person-in-the-loop in-browser editing, the operator-enabled public certificate and firewall rule, and the not-yet-recorded feature videos.

Page 2 — For software engineers

Release: codeserver-1.0.0-dev-0.0.3. **Release-gate aggregate:** 23/23 suites PASS, 0 FAIL — qa-results/run_all/20260701T151300Z-3182121/summary.txt. **Go gate:** 70 tests, -race, 81.8% coverage.

Architecture: Browser → Caddy (containerized rootless TLS edge, HTTP/3+Brotli, forward_auth) → helix-auth (Gin gate, host-native systemd --user, loopback :8081, non-root) + code-server (host-native as milosvasic, --auth none, loopback :8080).

Live-validated (PASS + captured evidence):

- **SSH-key challenge-response auth** — services/auth_gate/{challenge,verifier,cookie,server}.go; e2e_auth 5/5, challenges_auth CH1, helixqa_auth QA-002/004; Go 70 tests -race.
- **Login redirect** — server.go handleAuth (303→/login for HTML GET, 401 otherwise), commit 75e2d9b; TestAuthBrowserNavigationRedirectsToLogin + e2e_auth (B).
- **Fail-closed** — stress_chaos_auth 5/0/1 (live gate + code-server kill+recover), challenges_auth CH2, e2e_auth (E), helixqa_auth QA-003.
- **Cookie hardening** (__Host-+HMAC+HttpOnly+Secure+SameSite=Strict, regen on login) — cookie.go; security_auth 5/5.
- **DoS hardening** (rate-limit rightmost-XFF + CSRF double-submit + spawn ceiling) — ratelimit.go, server.go; load_auth 3/0/1, concurrency_auth 5/5, race 2/2.
- **CVE-2026-35414** server-pinned principal + key-type allow-list + OpenSSH floor — verifier.go, sshversion.go, config.go.
- **Host-native editor** — deploy/systemd/helix-code-server.service; editor loads (9 markers) authed 200 + X-Helix-User. Tarball install scripts/install-auth.sh (~/.local/lib, npm fallback, pinned 4.117.0).
- **Real-user env** — ssh-key git from terminal (CH3, git ls-remote 5 refs), .bashrc exports (CH4), Explorer defaults to \$PROJECTS_ROOT (CH5, convenience, not a jail).
- **Extension install+use** — tests/types/extensions_auth.sh + bank tests/banks/helixcode-extensions.yaml; 5/5 checks captured qa-results/tests/extensions_auth/20260701T162817Z-4080597 (Open VSX 200 → load → cleanup).
- **Login-form clipboard buttons** (commit 8351e4c) — services/auth_gate/assets/login_enhance.js
 - server.go; login_ui_auth.sh LIVE 3/3 = L1 served-page + L2 node unit 8/8 (qa-results/tests/login_ui_auth/20260701T184629Z-1069432/l2_node_unit.txt) + L3 §11.4.170 CDP pixel proof ok:true (render/labelled/non-overlap, copy→clipboard, paste→fill, 2-sig→picker, XSS-safe, no-auto-submit) qa-results/tests/login_ui_auth/

20260701T172810Z-584930/l3_visual.txt;

server_login_ui_test.go PASS.

- **VS Code Dark default theme** (commit 97a8c69) — deploy/code-server/settings.default.json = "Visual Studio Dark"; theme_default_auth.sh config 3/3 (qa-results/tests/theme_default_auth/20260701T184655Z-1071563) + theme_visual_auth.sh/ theme_visual_cdp.mjs §11.4.170 rendered-pixel proof: dark meanLum **44.3** (darkFrac 0.926, DOM vs-dark, bg rgb(37,37,38)) .../theme_visual_auth/20260701T183729Z-1053274 vs RED light-control meanLum **228.5** .../20260701T183816Z-1053829 (self-validated §11.4.107(10)).
- **Popular-extensions matrix** (commit 97a8c69) — extensions_popular_auth.sh; install proven live 4/5 .../extensions_popular_auth/20260701T181018Z-855046/p1_install.txt (+ direct 8-ext probe) + config round-trip OK; MS-proprietary honestly ABSENT from Open VSX (§11.4.112); install/persist steps network-gated (Open-VSX-unreachable runs install 0 = honest network-SKIP).
- **UI-driven marketplace journey** (commit 8351e4c) — extensions_ui_auth.sh (§11.4.48); headless Chromium opens Extensions view → searches Open VSX (rendered u3_search_results.png) → clicks the real Install button qa-results/tests/extensions_ui_auth/20260701T171228Z-232181 (5/1); install-completion is an honest operator_attended SKIP (headless cross-origin Open VSX net::ERR_ABORTED), autonomous on-disk proof = CLI sibling extensions_auth.
- **Caddy edge** — deploy/Caddyfile, deploy/caddy/Dockerfile (xcaddy-brotli); HTTP/3+Brotli, forward_auth, /proxy 403, X-Helix-User strip; security_auth 5/5, benchmark_auth 4/4. **Boot-persistent (LIVE):** rootless Quadlet deploy/quadlet/helixcode-caddy.container → helixcode-caddy.service active (running), Linger=yes, survives session crashes; edge answers HTTPS :52443.

Non-PASS states (honest §11.4.45):

- **PENDING — all Video-confirmation** (no real-use video corpus yet; every row's Video-confirmation is PENDING by design — PASS verdicts rest on the cited qa-results/** captured artefacts, not video, §11.4.6/§11.4.107). NOTE (§11.4.6): the §11.4.170 CDP pixel proofs (login buttons, dark theme) are cited to the runs where the driver reported ok:true/rc=0; the **latest** login_ui_auth L3 and theme_visual_auth runs honestly SKIPPED on a Chromium captureScreenshot timeout / workbench-never-rendered — a skipped CDP run is never counted as the pixel-proof PASS.
- **OPERATOR-ATTENDED** — in-browser editing (HCA-QA-UI-001 SKIP operator_attended).
- **OPERATOR-BLOCKED** — loopback firewall apply (harden-loopback.sh --apply, root); real public Let's Encrypt (public domain / DNS-01).

- **SKIP** — legacy container+password suites (integration, e2e, security, tls_letsencrypt, full_automation, concurrency, memory, benchmark, helixqa, challenges) → topology_unsupported, superseded by <name>_auth (no false-FAIL, §11.4.1).

Notes: code-server --auth none on loopback:8080 is a documented residual (auth enforced at gate/edge; harden-loopback.sh mitigates). DOCX exported via pandoc this release; no standing DOCX pipeline in tests/run_all_types.sh yet (tracked §11.4.60/§11.4.65 follow-up).